

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		Scales on y -axis multiples of 2, scale on x -axis multiples of 5 (1) [NB. Points must occupy at least half of each axis] 5 points plotted correctly to within <1 small square division (2) 4 correct (1) 3 or less (0) Best fit line drawn to within <1 small square division - line need not extend to the x -axis.(1)	1	1 1 1		4	4	4
		(ii)		5 (or accept reading from candidate's graph)		1		1	1	1
	(b)	(i)		Use of area under the graph or $W = \frac{1}{2} Fx$ (1) Substitution: $0.5 \times 15 \times 10^{-2} \times 12$ (1) $= 0.9$ (1) Only 90 (without working) on the answer line $\rightarrow$ 1 mark Joules / J (1) or N m [but not kJ etc.] Alternative: Use of $W = \text{mean force} \times \text{distance}$ (1) $= \frac{(0+12)}{2} \times 15 \times 10^{-2}$ (1) $= 0.9$ (1) Only 90 (without working) on the answer line $\rightarrow$ 1 mark Joules / J (1) or N m [but not kJ etc.]	1 1 1	1		4	3	3
		(ii)		[Work done =] KE = 0.9 [J] (1) (ecf from (i)) Manipulation and substitution: $v^2 = \frac{0.9 \text{ ecf}}{0.5 \times 55 \times 10^{-3}}$ [or euqiv] (1) $v = 5.7$ m/s too fast / doesn't meet guidelines (1) (ecf)		1 1	1	3	3	3
		(iii)		Plane has <u>less</u> KE (1)... ... because some energy [from spring system] 'wasted'/not transferred to plane [not: air resistance] (1)	2			2		1

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		(iv)		Momentum also takes account of mass / momentum = mv (1) Heavier object with same velocity [accept: speed] / 4 m/s has more momentum / even a lower velocity [accept: speed] could have more momentum = more dangerous (1)		2		2		
				Question 7 total	6	9	1	16	11	12